

Chapter 1

Definition

Definition 1.1 (Compactness).

Definition 1.2. Total Boundedness

Definition 1.3 (Cluster Point).

Definition 1.4. Sequential Compactness

Chapter 2

Fundamental Theorem of Compactness

Theorem 2.1 (Compactness $\Leftrightarrow$ Sequential Compactness).

Lemma 2.2 (Compactness $\Rightarrow$ Sequential Compactness). Let (X, d) be compact metric space, then X is sequentially compact.

Proof: Assume the contrary □

Lemma 2.3 (Compactness $\Leftarrow$ Sequential Compactness).

Proof:

$\Leftarrow$: Assume the contrary that X is not compact. G be a open cover of X with no finite subcover. □

Lemma 2.4 (Sequential Compactness $\Rightarrow$ Total Boundedness).

Lemma 2.5 (Sequential Compactness $\Rightarrow$ Completeness).

Theorem 2.6 (Cantor's Intersection Theorem). Let (X, d) be a complete metric space. Let $(V_n)_{n \in \mathbb{N}}$ where $V_1 \supseteq V_2 \supseteq \dots$ and V_i is closed and nonempty such that $d(V_n) = \sup_{x, y \in V_n} d(x, y) \rightarrow 0$. Then

$$\bigcap_{n \in \mathbb{N}} V_n$$

consists of one point.

Proof: At most one point one point: Suppose $x, y \in V_n$ for all n , then $d(x, y) \leq d(V_n) \rightarrow 0$ so $x = y$.

At least one point: Let $x_i \in V_i$ and create a sequence x_n . The sequence is Cauchy (nested sets gives us this). $n, m \geq N \Rightarrow V_n \cup V_m \subset V_N \Rightarrow d(x_n, x_m) \leq d(V_N) \rightarrow 0$. By Completeness, $x = \lim_{n \rightarrow \infty} x_n$ exists. If $n \geq m \Rightarrow x_n \in V_n \subset V_m$. Letting $n \rightarrow \infty$ while keeping m fixed and knowing V_m is closed, $x_n \rightarrow x \in V_m$. So x is in the Intersection. □

Chapter 2

References